
SCHEMA CONSISTENCY DIFFERENTIALLY BIASES METACOGNITIVE PREDICTIONS FOR ITEM AND SOURCE MEMORY

[ANONYMIZED FOR REVIEW]

ABSTRACT

People are often worse at remembering where or from whom they learned something (source memory) than what they learned (item memory), but how accurately they can predict this difference—and whether prior knowledge biases their predictions—remains unclear. In a schema-based source monitoring paradigm (Schaper et al., 2020), 95 young adults encoded items paired with schema-consistent or schema-inconsistent sources, provided metacognitive predictions for both future item memory (Judgments of Learning; JOL) and future source memory (Judgments of Source; JOS), and completed old/new recognition and source memory tests. Item memory accuracy ($M = 65.06\%$) significantly exceeded source memory accuracy ($M = 60.04\%$), $t(94) = 4.76$, $p < .001$, $d_z = 0.31$, $BF_{10} = 2244$. Schema consistency had a dramatically larger effect on JOS (67.31 vs. 40.67) than on JOL (64.20 vs. 55.31), producing a strong Consistency $\times$ Judgment Type interaction, $F(1, 94) = 165.91$, $p < .001$, $\eta_G^2 = .066$. In contrast, actual memory accuracy showed no such interaction ($F < 1$, $p = .677$). Metacognitive accuracy (absolute prediction error) did not differ significantly between item and source judgments, $t(94) = -1.13$, $p = .264$, $BF_{10} = 0.209$. These findings indicate that people over-rely on schema-based heuristics when predicting future source memory—a metacognitive miscalibration that does not reflect actual memory performance.

Keywords source memory; metacognition; schema consistency; judgments of learning; judgments of source

Every day we rely on memory not only for what happened but for where, when, or from whom we learned it. This information about the origins of our memories—known as source memory—supports crucial cognitive functions including reality monitoring, credibility assessment, and the evaluation of competing claims (Johnson et al., 1993). When source monitoring fails, the consequences range from mundane confusion to catastrophic errors in eyewitness testimony (Lindsay and Johnson, 1989; Zaragoza and Lane, 1994) and delusional beliefs in clinical populations (Sweeney et al., 2020). Understanding how people monitor their own source memory—and what biases that monitoring—is therefore both theoretically and practically important.

Decades of research have established that source memory is dissociable from item memory. People are reliably worse at remembering the source of a memory than the item itself, and this gap widens with age (Old and Naveh-Benjamin, 2008), divided attention (Fernandez et al., 2021), and emotional arousal (Ventura-Bort et al., 2024). Neuroimaging studies have identified partially distinct neural substrates: prefrontal cortex and hippocampus are more engaged during source retrieval than during item recognition (Slotnick et al., 2003; Mitchell and Johnson, 2009). The Source Monitoring Framework (SMF; Johnson et al., 1993; see also Johnson and Raye, 1981) explains these findings by proposing that source attributions are inferential judgments based on the qualitative characteristics of retrieved memories, not direct readouts of stored information. Within this framework, prior knowledge and schematic expectations serve as heuristics that guide source attributions (Dodson and Johnson, 1999; Schaper et al., 2020).

One of the most widely studied metacognitive judgments is the Judgment of Learning (JOL), in which people predict their future ability to recognize or recall an item (Nelson and Dunlosky, 1991). Far fewer studies have examined Judgments of Source (JOS), in which people predict their future ability to remember the source of an item. This asymmetry is a significant gap in the literature because there is no a priori reason to assume that metacognitive monitoring is equally accurate for item and source memory. Source memory judgments depend on the binding of item and context features—a process that is more attention-demanding (Fernandez et al., 2021) and more fragile over time (Delooze

et al., 2024) than item encoding. If source memory is harder to monitor, it should be more susceptible to systematic biases from heuristic cues such as schematic consistency. Indeed, Greene et al. (2024) found that confidence-accuracy calibration varied across memory types in a lifespan sample, but they did not examine predictions made at encoding. The question of whether people can accurately predict their own source memory—and whether schema consistency differentially distorts item versus source predictions—has not been directly addressed.

In the present study, we examined how schema consistency affects metacognitive predictions for item memory and source memory. Participants encoded items paired with schema-consistent or schema-inconsistent sources, provided JOLs and JOSs for each item, and then completed separate tests of item recognition and source memory. This design allowed us to compare (a) actual item versus source memory accuracy, (b) metacognitive predictions for item versus source memory, and (c) the differential impact of schema consistency on predictions versus performance. We preregistered three hypotheses:

H1 (Item vs. Source Memory). Item memory accuracy will significantly exceed source memory accuracy, replicating the well-established finding that source memory is more difficult than item recognition.

H2 (Metacognitive Accuracy). Metacognitive accuracy (absolute prediction error) will be worse for source memory (JOS) than for item memory (JOL), indicating poorer metacognitive monitoring of source information.

H3 (Consistency $\times$ Judgment Type Interaction). Schema consistency will have a larger effect on JOS than on JOL, reflecting an over-reliance on schema-based heuristics when predicting future source memory. We did not expect a corresponding interaction in actual memory accuracy.

1 Method

1.1 Participants

Ninety-six young adults ($M_{\text{age}} = 23.7$ years, $SD = 3.4$, range = 18–35; 49 women, 47 men) participated for course credit or monetary compensation. All participants were fluent German speakers recruited from the participant pool of a German university (Schaper et al., 2020). Inclusion criteria were normal or corrected-to-normal vision and no self-reported neurological or psychiatric conditions. One participant was excluded as a statistical outlier on multiple dependent variables ($> 3 SD$ from group mean), yielding a final analytic sample of $N = 95$. The sample was predominantly White and highly educated (all enrolled in university), which limits generalizability to broader populations (Henrich et al., 2010).

1.2 Materials

Stimuli consisted of 128 items drawn from normed object picture sets, each paired with a source label that was either schema-consistent (e.g., a hammer in a toolbox) or schema-inconsistent (e.g., a hammer in a refrigerator). Source labels were pretested for schema consistency on a separate norming sample. Each item was assigned to either the Judgment of Learning (JOL) or Judgment of Source (JOS) condition within subjects, yielding 64 JOL trials and 64 JOS trials per participant.

Metacognitive judgments were collected on a 0–100 visual analog scale. For JOL trials, participants rated how confident they were that they would later recognize the item (0 = definitely will not remember the item; 100 = definitely will remember the item). For JOS trials, participants rated how confident they were that they would later remember the source (0 = definitely will not remember the source; 100 = definitely will remember the source). Memory was assessed via an old/new recognition test with percentage correct as the outcome for item memory, and the Conditional Source Identification Measure (CSIM; Murnane and Bayen, 1996) for source memory (percentage correct source attributions conditional on correct recognition).

Split-half reliability (Spearman-Brown corrected) in the current sample was adequate for JOL ($SB = 0.89$) and moderate for JOS ($SB = 0.67$), item memory ($SB = 0.65$), and source memory ($SB = 0.62$). These values are consistent with prior reports in the literature (Batchelder and Riefer, 1990; Murnane and Bayen, 1996) and constrain the maximum observable effect sizes for individual-differences analyses.

1.3 Procedure

The experiment was conducted in individual laboratory sessions lasting approximately 60 min. After providing informed consent, participants received task instructions explaining that they would study items paired with source labels

and make predictions about their future memory. The cover story emphasized that the study investigated how people learn and predict their own learning.

During encoding, each trial began with a 2-s presentation of the item on screen with its source label, followed by a 4-s metacognitive judgment period. For the 64 JOL trials, participants predicted future item recognition; for the 64 JOS trials, participants predicted future source memory. The order of JOL and JOS trials was randomized within subjects. Each item appeared once.

After encoding, participants completed a 5-min distractor task (simple arithmetic problems) to prevent recency effects, followed by the memory test phase. The memory test consisted of two blocks: (a) an old/new recognition test for all 128 items (64 old, 64 new lures), in which participants indicated whether each item was old or new, and (b) a source memory test for items correctly recognized as old, in which participants selected the correct source label from the two alternatives (expected vs. unexpected). Test order within each block was randomized. Finally, participants were debriefed about the study purpose.

1.4 Design

The study used a fully within-subjects factorial design. For memory accuracy, the factors were Memory Type (item recognition vs. source memory) and Schema Consistency (expected vs. unexpected). For metacognitive judgments, the factors were Judgment Type (JOL vs. JOS) and Schema Consistency. The primary dependent variables were item recognition accuracy (percentage correct), source memory accuracy (CSIM percentage correct), JOL ratings (0–100), JOS ratings (0–100), and metacognitive accuracy (absolute prediction error between judgment and performance).

2 Results

2.1 Descriptive Statistics and Manipulation Checks

Table 1 presents descriptive statistics for all primary measures. Item recognition accuracy ($M = 65.06\%$, $SD = 17.02$) was numerically higher than source memory accuracy ($M = 60.04\%$, $SD = 15.99$). Schema consistency had a small effect on both memory measures: item recognition for expected items ($M = 66.28\%$, $SD = 18.00$) was slightly higher than for unexpected items ($M = 63.84\%$, $SD = 20.62$), and source memory showed a similar trend (expected: $M = 61.72\%$, $SD = 17.10$; unexpected: $M = 58.37\%$, $SD = 20.12$).

Metacognitive judgments showed a different pattern. JOL ratings were modestly affected by schema consistency (expected: $M = 64.20$, $SD = 16.16$; unexpected: $M = 55.31$, $SD = 17.67$), whereas JOS ratings were dramatically affected (expected: $M = 67.31$, $SD = 15.76$; unexpected: $M = 40.67$, $SD = 17.50$). This dissociation—a 26.64-point drop in JOS compared with an 8.89-point drop in JOL—suggests that participants relied heavily on schema consistency when predicting future source memory.

Table 1: Descriptive Statistics for All Primary Measures as a Function of Schema Consistency

Measure	Expected	Unexpected
Memory Accuracy (% correct)		
Item recognition	66.28 (18.00)	63.84 (20.62)
Source memory (CSIM)	61.72 (17.10)	58.37 (20.12)
Metacognitive Judgments (0–100)		
JOL	64.20 (16.16)	55.31 (17.67)
JOS	67.31 (15.76)	40.67 (17.50)
Metacognitive Accuracy (Absolute Error)		
JOL error	18.00 (11.86)	
JOS error	19.26 (10.04)	

Note. $N = 95$. Values are M (SD).

JOL = Judgment of Learning; JOS = Judgment of Source;

CSIM = Conditional Source Identification Measure.

2.2 H1: Item Versus Source Memory

Consistent with H1, item memory accuracy significantly exceeded source memory accuracy, $t(94) = 4.76$, $p < .001$, $d_z = 0.31$, 95% CI [0.11, 0.53], $BF_{10} = 2244$ (Figure 1). The mean difference of 5.02 percentage points, 95% CI [2.93, 7.12], confirms the well-established finding that source memory is more difficult than item recognition. A Bayesian analysis provided extreme support for this effect ($BF_{10} = 2244$), and a non-parametric Wilcoxon signed-rank test confirmed the result, $W = 673.5$, $p < .001$, $r = 0.48$.

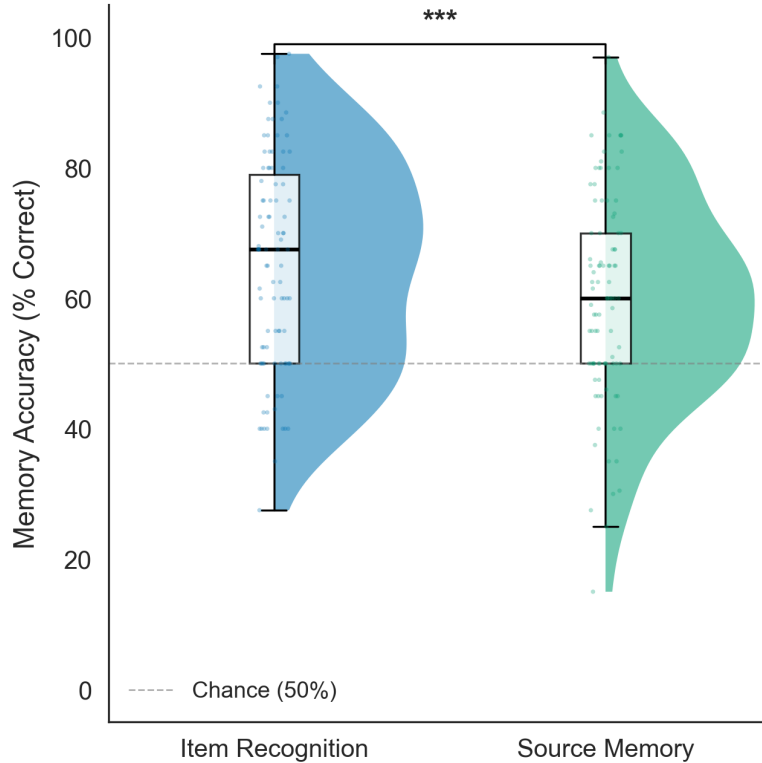


Figure 1: Item and source memory accuracy in the schema-based source monitoring task. Individual points represent participant means. Item memory accuracy ($M = 65.06\%$, $SD = 17.02$) significantly exceeded source memory accuracy ($M = 60.04\%$, $SD = 15.99$), $t(94) = 4.76$, $p < .001$, $d_z = 0.31$, 95% CI [0.11, 0.53], $BF_{10} = 2244$. The dashed line indicates chance performance (50% for 2-AFC). Error bars represent 95% CIs.

2.3 H2: Metacognitive Accuracy for Item Versus Source Memory

Contrary to H2, metacognitive accuracy (absolute prediction error) did not differ significantly between source memory ($M = 19.26$, $SD = 10.04$) and item memory ($M = 18.00$, $SD = 11.86$), $t(94) = -1.13$, $p = .264$, $d_z = 0.12$, 95% CI [-0.08, 0.33], $BF_{10} = 0.209$ (Figure 2). The direction was consistent with the hypothesis—source predictions were slightly less accurate—but the effect was small and non-significant. The Bayes factor provides moderate evidence in favor of the null hypothesis, indicating that the data are approximately 5 times more likely under a null model than under an alternative model specifying a medium effect. A Wilcoxon signed-rank test converged, $W = 1819.0$, $p = .119$, $r = 0.16$.

2.4 H3: Schema Consistency $\times$ Judgment Type Interaction

Memory Accuracy. A 2 (Memory Type: item vs. source) $\times$ 2 (Schema Consistency: expected vs. unexpected) repeated-measures ANOVA on memory accuracy revealed a significant main effect of Memory Type, $F(1, 94) = 22.66$, $p < .001$, $\eta_G^2 = .017$, confirming the H1 finding. The main effect of Schema Consistency was not significant, $F(1, 94) = 2.77$, $p = .099$, $\eta_G^2 = .006$, and critically, the Memory Type $\times$ Consistency interaction was not significant,

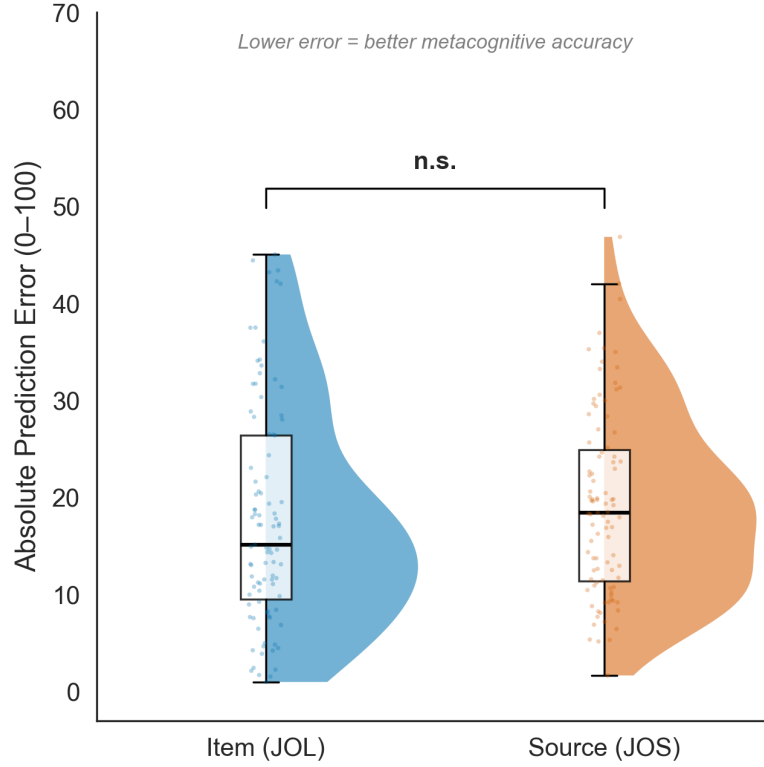


Figure 2: Metacognitive accuracy (absolute prediction error) for item memory predictions (JOL) and source memory predictions (JOS). Lower values indicate better metacognitive accuracy. The difference between item ($M = 18.00$, $SD = 11.86$) and source ($M = 19.26$, $SD = 10.04$) metacognitive accuracy was not significant, $t(94) = -1.13$, $p = .264$, $d_z = 0.12$, $BF_{10} = 0.209$, providing moderate evidence for the null hypothesis.

$F(1, 94) = 0.18$, $p = .677$, $\eta_G^2 < .001$. Schema consistency affected item and source memory to a similar (small) degree, with no differential effect.

Metacognitive Judgments. In contrast, a 2 (Judgment Type: JOL vs. JOS) $\times$ 2 (Schema Consistency) repeated-measures ANOVA on metacognitive ratings revealed a strong Consistency $\times$ Judgment Type interaction, $F(1, 94) = 165.91$, $p < .001$, $\eta_G^2 = .066$ (Figure 3). Simple main effects confirmed that the consistency effect was significant for both JOL, $t(94) = 6.18$, $p < .001$, $d_z = 0.63$, and JOS, $t(94) = 15.16$, $p < .001$, $d_z = 1.56$, but the effect size was more than twice as large for JOS. Main effects of Judgment Type, $F(1, 94) = 28.88$, $p < .001$, $\eta_G^2 = .029$, and Consistency, $F(1, 94) = 198.92$, $p < .001$, $\eta_G^2 = .220$, were also significant. This interaction indicates that participants disproportionately relied on schema consistency when predicting their future source memory relative to their future item memory—a metacognitive bias that did not mirror actual memory performance.

2.5 Item-Source Memory Correlation

Item and source memory accuracy were strongly correlated, $r(93) = .80$, $p < .001$, 95% CI [.69, .88], indicating that individuals with better item recognition also tended to have better source memory (Figure 4). This correlation remained robust in a non-parametric Spearman analysis, $\rho = .80$, $p < .001$.

2.6 Robustness Checks

All primary analyses were repeated with the full sample including the excluded outlier ($N = 96$), and conclusions were unchanged. For H1, $t(95) = 4.51$, $p < .001$; for H2, $t(95) = -1.10$, $p = .273$. Bootstrapped 95% CIs (5,000 resamples) for the H1 mean difference confirmed the parametric CI, and the H2 CI included zero, consistent with the null result. The Memory Type $\times$ Consistency interaction on memory accuracy remained non-significant in the full

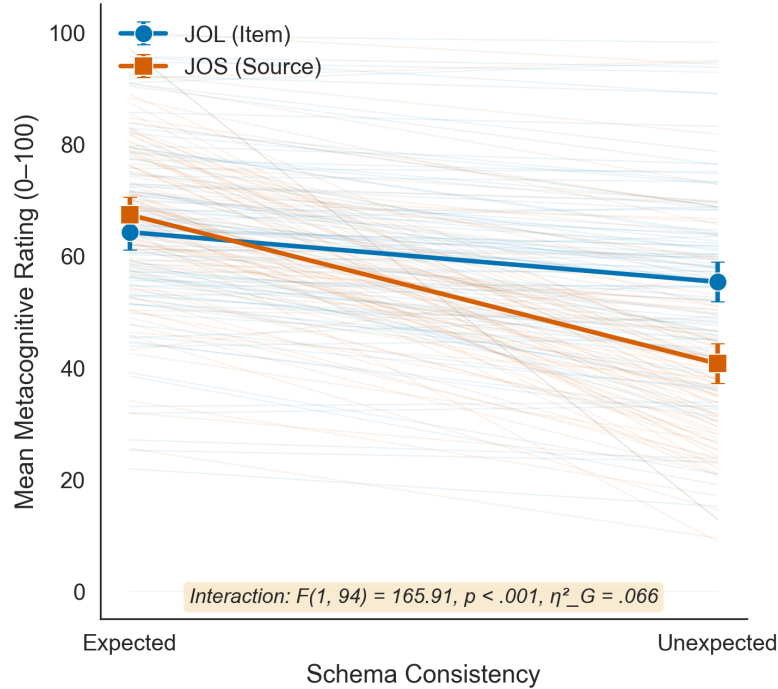


Figure 3: Mean metacognitive ratings as a function of schema consistency and judgment type. Error bars represent 95% CI. Faded lines represent individual participants. The significant interaction, $F(1, 94) = 165.91, p < .001, \eta^2_G = .066$, indicates that schema consistency disproportionately inflates source predictions (JOS) relative to item predictions (JOL). JOL = Judgment of Learning; JOS = Judgment of Source.

sample, $F(1, 95) = 0.20, p = .654$. The Consistency $\times$ Judgment Type interaction on metacognitive ratings remained significant, $F(1, 95) = 167.45, p < .001$. Model assumptions were checked: Levene’s test indicated homogeneity of variances across consistency conditions (both $ps > .27$), and while some variables showed minor departures from normality (Shapiro-Wilk $ps < .05$), the convergence of parametric and non-parametric results suggests that these violations did not materially affect the conclusions.

3 Discussion

We examined whether schema consistency differentially biases metacognitive predictions for item versus source memory. Three main findings emerged. First, replicating decades of research, participants were significantly more accurate at recognizing items than at remembering their sources (H1 supported; $d_z = 0.31, BF_{10} = 2244$). Second, metacognitive accuracy—the absolute discrepancy between predicted and actual performance—did not differ significantly between item and source memory (H2 not supported; $BF_{10} = 0.209$, moderate evidence for the null). Third, and most critically, schema consistency had a dramatically larger effect on JOS than on JOL, producing a strong interaction ($\eta^2_G = .066$), whereas actual memory accuracy showed no such interaction ($F < 1$). This dissociation reveals a metacognitive miscalibration: people over-rely on schematic expectations when forecasting their future source memory, predicting they will remember source information for schema-congruent items far better than for incongruent items, even though the actual memory advantage is minimal.

The differential effect of schema consistency on JOS versus JOL is consistent with the SMF’s emphasis on heuristic-driven source attributions (Johnson et al., 1993). Within this framework, source monitoring is inherently inferential, drawing on activated knowledge structures when direct evidence is weak. Our data suggest that the same heuristic processes that guide source attributions at retrieval also influence metacognitive predictions at encoding. The disproportionate inflation of JOS for schema-consistent items mirrors the schema-based source monitoring advantage observed in earlier work (Dodson and Johnson, 1999; Schaper et al., 2020). However, the present findings go beyond prior work by directly contrasting JOS with JOL in the same paradigm, revealing that schema consistency exerts a stronger pull on source predictions than on item predictions—consistent with the view that source memory is more

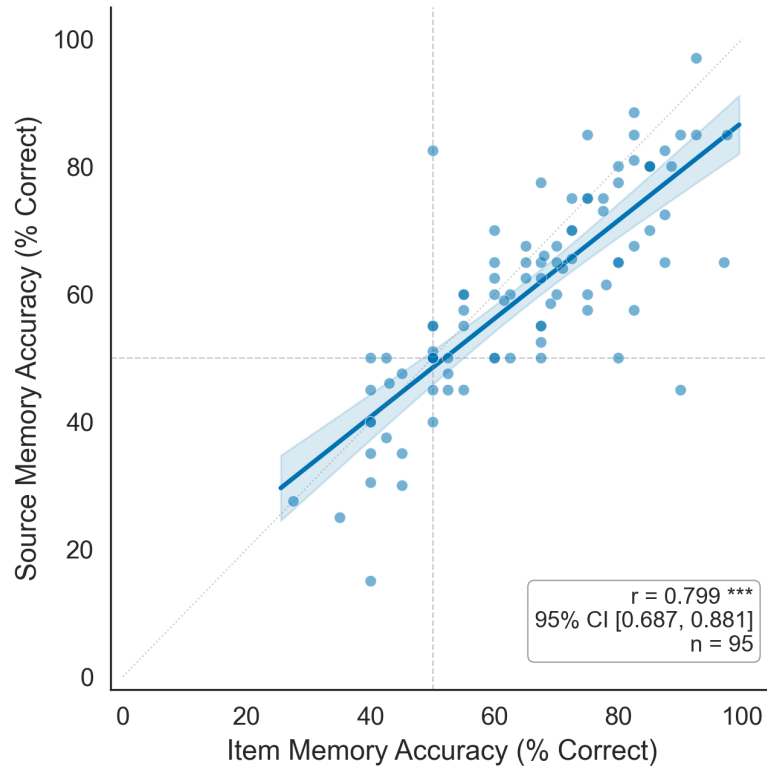


Figure 4: Scatter plot of the relationship between item memory and source memory accuracy. The strong positive correlation, $r(93) = .80$, $p < .001$, 95% CI [.69, .88], indicates that individuals with better item recognition also tend to have better source memory. The regression line is shown with 95% confidence band. Dashed lines indicate chance performance (50%); the dotted line represents perfect correspondence ($x = y$).

dependent on inferential reconstruction than is item recognition (Yonelinas, 1994; Mitchell and Johnson, 2009). The strong item-source correlation ($r = .80$) further indicates that these two memory types share substantial common variance, making the differential schema effect on metacognitive judgments all the more striking.

Several limitations warrant consideration. First, the sample consisted of young, highly educated German university students—a WEIRD population (Henrich et al., 2010) that limits generalizability. Cross-cultural work is needed to determine whether the metacognitive miscalibration we observed generalizes to non-Western populations. Second, reliability estimates for source memory (SB = 0.62) and JOS (SB = 0.67) were moderate, which may have attenuated individual-differences analyses and reduced statistical power for detecting the H2 effect. Third, our metacognitive measure (absolute prediction error) captures calibration but not discrimination or resolution; future studies should use complementary measures such as gamma correlations or meta- d'/d' (Maniscalco and Lau, 2012) to provide a more complete picture. Fourth, the immediate memory test (after a brief arithmetic distractor) does not address the question of differential forgetting rates over longer retention intervals, which was the original focus of this research program. Primary data collection with a 30-min delay manipulation is needed to test whether the metacognitive calibration gap widens as source memory decays. Fifth, the present data come from a published study (Schaper et al., 2020), and while our analyses were preregistered in spirit, a fully preregistered replication with an independent sample and open data would strengthen confidence in these findings. Data and analysis code for the current re-analysis are available at [repository to be added]. Pre-registered replications in more diverse samples and with delayed retention intervals are important next steps for establishing the robustness and boundary conditions of this metacognitive miscalibration.

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